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### High-K capacitor dielectric having a metal oxide area comprising boron, electrical device and semiconductor apparatus including the same

#### Abstract

Provided is a semiconductor device including a lower electrode, an upper electrode isolated from direct contact with the lower electrode, and a dielectric layer between the lower electrode and the upper electrode, the dielectric layer comprising a first metal oxide area, a second metal oxide area, and a third metal oxide area. The third metal oxide area is between the first metal oxide area and the second metal oxide area, and includes boron and one or more metal elements selected from aluminum (Al), magnesium (Mg), silicon (Si), or beryllium (Be). In the third metal oxide area, a content of boron (B) is less than or equal to a content of the metal elements of Al, Mg, Si, and/or Be.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation of U.S. application Ser. No. 17/146,894, filed on Jan. 12, 2021, which is based on and claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2020-0111687, filed on Sep. 2, 2020, in the Korean Intellectual Property Office, the disclosures of each of which are incorporated by reference herein in its entirety.

### BACKGROUND

#### 1. Field

(1) The disclosure relates to an electrical device and a semiconductor apparatus including the electrical device.

#### 2. Description of Related Art

(2) As electronic apparatuses are down-scaled, space occupied by electrical devices in electronic apparatuses also decreases. Accordingly, with a decrease in the size of electrical devices, such as capacitors, a decrease in the thickness of a dielectric layer included in the electrical devices is also required. However, in this case, as some leakage current is greatly generated through the dielectric layer of a capacitor, driving of a device may become difficult.

### SUMMARY

(3) Provided is an electrical device having a high capacitance and a low leakage current value, and a semiconductor apparatus including the electrical device.

(4) Provided is an electrical device having a dielectric layer including three or more metal oxide area.

(5) Provided is an electrical device having a dielectric layer including a metal oxide layer including three or more types of metal elements.

(6) Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments of the disclosure.

(7) According to an example embodiment, an electrical device includes a lower electrode, an upper electrode isolated from direct contact with the lower electrode, and a dielectric layer between the lower electrode and the upper electrode, the dielectric layer including a first metal oxide area, a second metal oxide area, and a third metal oxide area.

(8) The first metal oxide area and the second metal oxide area each independently may have a dielectric constant of 20 or more and 70 or less.

(9) The third metal oxide area may be between the first metal oxide area and the second metal oxide area and may including boron (B) and one or more metal elements selected from aluminum (Al), magnesium (Mg), silicon (Si), or beryllium (Be).

(10) The dielectric layer may include at least one of a first metal oxide layer in the first metal oxide area, a second metal oxide layer in the second metal oxide area, or a third metal oxide layer in the third metal oxide area

(11) The first metal oxide area, the second metal oxide area, and the third metal oxide area are sequentially arranged in a thickness direction of the dielectric layer. In the third metal oxide area, the content of B may be less than or equal to the content of a metal element such as Al, Mg, Si, and/or Be.

(12) The third metal oxide area may further include one or more metal elements selected from Ca,

Sr, Ba, Sc, Y, La, Ti, Hf, Zr, Nb, Ta, Ce, Pr, Nd, Gd, Dy, Yb, Pb, Zn, or Lu.

(13) The third metal oxide area may include a metal oxide represented by  $AB_{\text{sub.a}}C_{\text{sub.1-a}}O$ . A may be one or more elements selected from Ca, Sr, Ba, Sc, Y, La, Ti, Hf, Zr, Nb, Ta, Ce, Pr, Nd, Gd, Dy, Yb, Pb, Zn, or Lu, B may be boron (B), C may be one or more elements selected from Al, Mg, Si, and Be, and “a” may be greater than 0.00 and equal to or less than 0.50.

(14) The first metal oxide area may be adjacent to the lower electrode, and the thickness of the first metal oxide layer may be 40% or more of the total thickness of the dielectric layer.

(15) A thickness of the first metal oxide area may be 10 Å or more and 50 Å or less; a thickness of the second metal oxide area may be 10 Å or more and 50 Å or less; and/or a thickness of the third metal oxide area may be 10 Å or more and 50 Å or less. A thickness of the dielectric layer may be 20 Å or more and 100 Å or less. The ratio of a thickness of the third metal oxide area with respect to a thickness of the first metal oxide area may be 0.3 or more and less than 1.0.

(16) The dielectric layer may include a fourth metal oxide area between the upper electrode and the second metal oxide area, the fourth metal oxide area comprising one or more metal elements selected from Al, Mg, Si, and Be. The fourth metal oxide area may comprise a smaller content of boron (B) than a boron content of the third metal oxide. The fourth metal oxide area may, for example, not include boron. A thickness of the fourth metal oxide layer is 5 Å or more and 50 Å or less.

(17) The dielectric layer may be configured such that, when a voltage of 1.0 V is applied, a leakage current value is  $1.0 \times 10^{-4}$  A/cm<sup>2</sup> or less.

(18) At least one of the upper electrode and the lower electrode may comprise a metal nitride represented by  $MM'N$ , wherein M is a metal element, M' is a doping element, and N is nitrogen.

(19) The electrical device may further comprising an interface layer between the dielectric layer and at least one of the upper electrode and the lower electrode. The interface layer may comprise a transition metal oxide.

(20) A semiconductor apparatus may include the electrical device; and a field-effect transistor electrically connected to the electrical device. The field-effect transistor may include a semiconductor layer comprising a source and a drain; a gate dielectric layer on the semiconductor layer; and a gate electrode on the gate dielectric layer.

(21) According to another embodiment, a semiconductor apparatus includes a lower electrode, an upper electrode isolated from direct contact with the lower electrode, and a dielectric layer between the lower electrode and the upper electrode and including one or more first metal elements selected from Ca, Sr, Ba, Sc, Y, La, Ti, Hf, Zr, Nb, Ta, Ce, Pr, Nd, Gd, Dy, Yb, Pb, Zn, or Lu, one or more second metal elements selected from Al, Mg, Si, or Be, and boron (B).

(22) The content of the boron (B) in the dielectric layer may be less than or equal to the content of the one or more second metal element. The dielectric layer may comprise a metal oxide represented by  $AB_{\text{sub.a}}C_{\text{sub.1-a}}O$ , wherein A represents the one or more first metal elements, B represents the boron (B), C represents the one or more second metal element, and a is less than or equal to 0.50.

(23) The content of the boron (B) may be greater than 0.0 at % and equal to or less than 3.0 at % with respect to the total of metal elements of the dielectric layer. The content of the one or more first metal elements may be 92 at % or more and less than 100 at % with respect to the total of metal elements of the dielectric layer. The content of the second metal element may be greater than 0.0 at % and equal to or less than 5.0 at % with respect to the total of metal elements of the dielectric layer.

(24) The boron (B) may have a concentration gradation in a thickness direction of the dielectric layer. The dielectric layer may include, sequentially in a thickness direction, sequentially in a thickness direction, a lower surface facing the lower electrode, an inner area above the lower surface, and an upper surface above the inner area and facing the upper electrode. The boron (B) may have a maximum concentration at a position away from the lower electrode by 40% or more

and 90% or less of a thickness of the dielectric layer. A thickness of the dielectric layer is 20 Å or more and 100 Å or less.

(25) The dielectric layer may be configured such that, when a voltage of 1.0 V is applied, a leakage current value may be  $1.0 \times 10^{-4}$  A/cm<sup>2</sup> or less.

(26) A semiconductor apparatus may include the electrical device; and a field-effect transistor electrically connected to the electrical device. The field-effect transistor may include a semiconductor layer comprising a source and a drain; a gate dielectric layer on the semiconductor layer; and a gate electrode on the gate dielectric layer.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

(2) FIGS. 1, 2, and 3 are schematic views of an electrical device according to some example embodiments;

(3) FIG. 4A illustrates a circuit configuration of a memory cell of a memory device including a semiconductor device and a capacitor;

(4) FIG. 4B is a schematic view of a semiconductor apparatus according to an example embodiment;

(5) FIG. 5 is a layout diagram of a semiconductor apparatus according to an example embodiment;

(6) FIGS. 6 and 7 are cross-sectional views taken along line A-A' of the semiconductor apparatus of FIG. 5;

(7) FIGS. 8 and 9 are conceptual views schematically showing device architectures applicable to an electronic apparatus according to some embodiments;

(8) FIGS. 10A to 10H are views conceptually illustrating the respective operations of a method of manufacturing a metal nitride film according to some example embodiments; and

(9) FIG. 11 is a structure of a trench capacitor-type dynamic random access memory (DRAM) according to an example embodiment.

### DETAILED DESCRIPTION

(10) Reference will now be made in detail to some example embodiments, examples of which are illustrated in the accompanying drawings, wherein like reference numerals refer to like elements throughout. In this regard, the present embodiments may have different forms and should not be construed as being limited to the descriptions set forth herein. Accordingly, the embodiments are merely described below, by referring to the figures, to explain aspects. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

Expressions such as “at least one of,” when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list.

(11) The terms used in the specification are merely used to describe particular embodiments, and are not intended to limit the disclosure. when a constituent element is disposed “above” or “on” to another constituent element, the constituent element may include not only an element directly contacting on the upper/lower/left/right sides of the other constituent element, but also an element disposed above/under/left/right the other constituent element in a non-contact manner.

(12) An expression used in a singular form in the specification also includes the expression in its plural form unless clearly specified otherwise. Also, terms such as “include” or “comprise” may be construed to denote a certain characteristic, number, step, operation, constituent element, or a combination thereof, but may not be construed to exclude the existence of or a possibility of addition of one or more other characteristics, numbers, steps, operations, constituent elements, or

combinations thereof.

(13) Terms such as “first” and “second” are used herein merely to describe a variety of constituent elements, but the constituent elements are not limited by the terms. Such terms are used only for the purpose of distinguishing one constituent element from another constituent element.

Furthermore, terms such as “~portion,” “~unit,” “~module,” and “~block” stated in the specification may signify a unit to process at least one function or operation and the unit may be embodied by hardware, software, or a combination of hardware and software.

(14) Hereinafter, the disclosure will be described in detail by explaining some example embodiments of the disclosure with reference to the attached drawings. As noted above, like reference numerals in the drawings denote like elements, and in the drawings, the size of each constituent element (width, thickness, and the like of a layer, an area, and the like) may be exaggerated for clarity and convenience of explanation. Embodiments described below are merely examples, and various modifications are possible from the embodiments.

(15) According to one aspect, an electrical device having a low leakage current and a high capacitance may be provided. The electrical device may be a capacitor.

(16) FIG. 1 is a schematic view of a capacitor **1** according to an example embodiment. Referring to FIG. 1, the capacitor **1** may include a lower electrode **100**, an upper electrode **200** arranged apart from the lower electrode **100**, and a dielectric layer **300** arranged between the lower electrode **100** and the upper electrode **200**.

(17) The lower electrode **100** may be disposed on a substrate (not shown). The substrate may be a part of a structure for supporting the capacitor **1** and/or a part of a device connected to the capacitor **1**. The substrate may include a semiconductor material pattern, an insulating material pattern, and/or a conductive material pattern. The substrate may include, for example, a substrate **11'**, a gate stack **12**, an interlayer insulating layer **15**, a contact structure **20'**, and/or a bit line structure **13**, which are illustrated below in FIGS. 6 and 7. Furthermore, the substrate may include, for example, a semiconductor material such as silicon (Si), germanium (Ge), silicon germanium (SiGe), silicon carbide (SiC), gallium arsenic (GaAs), indium arsenic (InAs), indium phosphide (InP), and/or the like, an insulating material such as a silicon oxide, a silicon nitride, a silicon oxynitride, and the like, and/or a conductive material such as a metal.

(18) The upper electrode **200** may be arranged apart from the lower electrode **100** such that the upper electrode **200** and the lower electrode **100** face each other. The lower electrode **100** and/or the upper electrode **200** each may independently include metal, metal nitride, metal oxide, or a combination thereof. For example, the lower electrode **100** and/or the upper electrode **200** each may independently include metal (such as ruthenium (Ru), titanium (Ti), tantalum (Ta), niobium (Nb), iridium (Ir), molybdenum (Mo), tungsten (W), platinum (Pt), and the like), a conductive metal nitride (such as titanium nitride (TiN), tantalum nitride (TaN), niobium nitride (NbN), molybdenum nitride (MoN), cobalt nitride (CoN), tungsten nitride (WN), and the like), and/or a conductive metal oxide (such as platinum oxide (PtO), iridium oxide (IrO.sub.2), ruthenium oxide (RuO.sub.2), strontium ruthenium oxide (SrRuO.sub.3), a barium strontium ruthenium oxide ((Ba,Sr)RuO.sub.3), calcium ruthenium oxide (CaRuO.sub.3), lanthanum strontium cobalt oxide ((La,Sr)CoO.sub.3), indium tin oxide (ITO), and the like).

(19) For example, the lower electrode **100** and/or the upper electrode **200** each may independently include a metal nitride represented by MM'N. In this case, M may denote a metal element, M' may denote an element different from M, and N may denote nitrogen. For example, the metal nitride may include an MN metal nitride doped with the element M'. M may be one, two, and/or more elements selected from among Be, B, Na, Mg, Al, Si, K, Ca, Sc, Ti, V, Cr, Mn, Fe, Co, Ni, Cu, Zn, Ga, Ge, Rb, Sr, Y, Zr, Nb, Mo, Tc, Ru, Rh, Pd, Ag, Cd, In, Sn, Sb, Te, Cs, Ba, La, Ce, Pr, Nd, Pm, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, Lu, Hf, Ta, W, Re, Os, Ir, Pt, Au, Hg, Tl, Pb, Bi, Po, Fr, Ra, Ac, Th, Pa, and U. M' may be one, two, and/or more elements selected from among H, Li, Be, B, N, O, Na, Mg, Al, Si, P, S, K, Ca, Sc, Ti, V, Cr, Mn, Fe, Co, Ni, Cu, Zn, Ga, Ge, As, Se, Rb, Sr, Y,

Zr, Nb, Mo, Tc, Ru, Rh, Pd, Ag, Cd, In, Sn, Sb, Te, Cs, Ba, La, Ce, Pr, Nd, Pm, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, Lu, Hf, Ta, W, Re, Os, Jr, Pt, Au, Hg, Tl, Pb, Bi, Po, Fr, Ra, Ac, Th, Pa, and U. When a composition ratio of M, M', and N of a metal nitride MM'N is  $x:y:z$  (e.g.,  $M_{\text{sub}.x}M'_{\text{sub}.y}N_{\text{sub}.z}$ )  $0 \leq x \leq 2$ ,  $0 \leq y \leq 2$ , and  $0 < z \leq 4$ , where at least one of  $x$  and  $y$  is not 0. In the case wherein M' is nitrogen (N) (e.g., wherein M' and N are the same) the composition of the lower electrode **100** and/or the upper electrode **200** may be represented by  $M_{\text{sub}.x}N_{\text{sub}.z}$  with  $0 < x \leq 2$  and  $0 < z \leq 6$ .

(20) The lower electrode **100** and/or the upper electrode **200** each may independently be a single material layer and/or a stacked structure of a plurality of material layers. For example, the lower electrode **100** and/or the upper electrode **200** each may independently be a single layer of TiN and/or a single layer of NbN. Alternatively, the lower electrode **100** and/or the upper electrode **200** may have a stacking structure including a first electrode layer including TiN and a second electrode layer including NbN.

(21) The dielectric layer **300** may include a first metal oxide layer **310**, a second metal oxide layer **330**, and a third metal oxide layer **320**. The first metal oxide layer **310** may be arranged close to the lower electrode **100**, the second metal oxide layer **330** may be arranged apart from the first metal oxide layer **310** such that the first metal oxide layer **310** and the second metal oxide layer **330** face each other, and the third metal oxide layer **320** may be arranged between the first metal oxide layer **310** and the second metal oxide layer **330**. For example, the first metal oxide layer **310**, the third metal oxide layer **320**, and the second metal oxide layer **330** may be sequentially arranged in a thickness direction of the dielectric layer **300** (e.g., in a vertical direction when viewed in a cross-section).

(22) The first metal oxide layer **310** and/or the second metal oxide layer **330** may have a high dielectric constant. For example, the first metal oxide layer **310** and/or the second metal oxide layer **330** each may independently have a dielectric constant of 20 or more and/or 70 or less. The first metal oxide layer **310** and/or the second metal oxide layer **330** may have paraelectric properties. The first metal oxide layer **310** and/or the second metal oxide layer **330** each may independently include one, two, and/or more metal selected from among Ca, Sr, Ba, Sc, Y, La, Ti, Hf, Zr, Nb, Ta, Ce, Pr, Nd, Gd, Dy, Yb, Pb, Zn and Lu. For example, the first metal oxide layer **310** and/or the second metal oxide layer **330** each may independently include a hafnium oxide ( $\text{HfO}_{\text{sub}.2}$ ), a hafnium silicon oxide ( $\text{HfSiO}_{\text{sub}.4}$ ), a lanthanum oxide ( $\text{La}_{\text{sub}.2}\text{O}_{\text{sub}.3}$ ), a lanthanum aluminum oxide ( $\text{LaAlO}_{\text{sub}.3}$ ), a zirconium oxide ( $\text{ZrO}_{\text{sub}.2}$ ), a hafnium zirconium oxide ( $\text{HfZrO}_{\text{sub}.2}$ ), a zirconium silicon oxide ( $\text{ZrSiO}_{\text{sub}.4}$ ), a tantalum oxide ( $\text{Ta}_{\text{sub}.2}\text{O}_{\text{sub}.5}$ ), a titanium oxide ( $\text{TiO}_{\text{sub}.2}$ ), a strontium titanium oxide ( $\text{SrTiO}_{\text{sub}.3}$ ), an yttrium oxide ( $\text{Y}_{\text{sub}.2}\text{O}_{\text{sub}.3}$ ), an aluminum oxide ( $\text{Al}_{\text{sub}.2}\text{O}_{\text{sub}.3}$ ), a cerium oxide ( $\text{CeO}_{\text{sub}.2}$ ), a lead scandium tantalum oxide ( $\text{PbScTaO}_{\text{sub}.3}$ ), a lead zinc niobate ( $\text{PbZnNbO}_{\text{sub}.3}$ ), and the like. Furthermore, the first metal oxide layer **310** and/or the second metal oxide layer **330** each may independently include a metal oxynitride (such as an aluminum oxynitride (AlON), a zirconium oxynitride (ZrON), a hafnium oxynitride (HfON), a lanthanum oxynitride (LaON), a yttrium oxynitride (YON), and the like), a silicate (such as  $\text{ZrSiON}$ ,  $\text{HfSiON}$ ,  $\text{YSiON}$ ,  $\text{LaSiON}$ , and the like), and/or an aluminate (such as  $\text{ZrAlON}$ ,  $\text{HfAlON}$ , and the like).

(23) The third metal oxide layer **320** may include boron (B) and at least one, two, and/or more metal elements selected from aluminum (Al), magnesium (Mg), silicon (Si), and/or beryllium (Be). The third metal oxide layer **320** may be a leakage current reducing layer that reduces and/or blocks a leakage current flowing in the capacitor **1**. For example, a capacitor including the first metal oxide layer **310** and/or the second metal oxide layer **330**, without the third metal oxide layer **320**, may be difficult to operate because of an excessive leakage current flows in the capacitor. When a metal oxide layer including one, two, and/or more of Al, Mg, Si, and Be, but not including B, is provided between the first metal oxide layer **310** and the second metal oxide layer **330**, a leakage current in the capacitor may decrease, but the capacitance of the capacitor may be lowered. In



contrast, when a metal oxide layer including boron (B) and one, two, and/or more of Al, Mg, Si, and/or Be, is provided between the first metal oxide layer **310** and the second metal oxide layer **330**, a leakage current in the capacitor may be reduced and simultaneously the capacitance of the capacitor may be maintained and/or a decreasing width may be reduced. Without being limited to a specific theory, the lower leakage current in the capacitor may be due to a higher crystallinity in the boron including third metal oxide layer **320** compared to a metal oxide layer including Al, Mg, Si, and/or Be without boron, as boron is less likely to inhibit crystal growth in the metal oxide layer than Al, Mg, Si, Be, or the like. For example, the third metal oxide layer **320** may have a higher crystallinity than the first metal oxide layer **310** and/or the second metal oxide layer **330**.

(24) Furthermore, the third metal oxide layer **320** may further include one, two, and/or more metal elements selected from Ca, Sr, Ba, Sc, Y, La, Ti, Hf, Zr, Nb, Ta, Ce, Pr, Nd, Gd, Dy, Yb, Pb, Zn, and/or Lu.

(25) In the third metal oxide layer **320**, the content of boron (B) may be less than or equal to the content of the one, two, and/or more of Al, Mg, Si, and/or Be. For example, the content of B in the third metal oxide layer **320** may be 0.95 or less, 0.90 or less, 0.80 or less, 0.75 or less, and/or 0.70 or less, and/or 0.01 or more, 0.05 or more, 0.10 or more, 0.15 or more, 0.20 or more, 0.25 or more, and/or 0.30 or more than the content of the one, two, and/or more metal elements selected from Al, Mg, Si, and/or Be.

(26) For example, the third metal oxide layer **320** may include a metal oxide represented by  $AB.sub.aC.sub.1-aO$ , wherein A may denote the one, two, and/or more metal elements selected from Ca, Sr, Ba, Sc, Y, La, Ti, Hf, Zr, Nb, Ta, Ce, Pr, Nd, Gd, Dy, Yb, Pb, Zn, and Lu, B denotes boron, and C may denote the one, two, and/or more metal elements selected from Al, Mg, Si, and/or Be. A ratio (a) of B and the metal element C may be greater than 0.00, for example, 0.10 or more, 0.15 or more, 0.20 or more, and/or 0.50 or less, and/or 0.45 or less. Furthermore, the metal element A, the metal element C, and/or B in the third metal oxide layer **320** may be included in appropriate amounts according to a desired dielectric constant of a dielectric layer, a desired leakage current value of a capacitor, and/or the like. For example, the content of B in the third metal oxide layer **320** may be greater than 0.0 at %, 0.3 at % or more, 0.5 at % or more, 0.7 at % or more, 1.0 at % or more, 1.5 at % or more, and/or 2.0 at % or more, and/or 10.0 at % or less, 7.0 at % or less, 5.0 at % or less, 4.0 at % or less, 3.0 at % or less, 2.5 at % or less, 2.0 at % or less, and/or 1.5 at % or less, with respect to a total content of the metal elements in the third metal oxide layer **320**. The content of the metal element A, in the third metal oxide layer **320**, may be, for example, 80 at % or more, 85 at % or more, and/or 90 at % or more, and/or less than 100 at %, 98 at % or less, and/or 96 at % or less, with respect to the total content of metal elements in the third metal oxide layer **320**. Furthermore, in the metal oxide represented by  $AB.sub.aC.sub.1-aO$ , an element ratio among the metal element A, B, and the metal element C may be determined according to the content of each metal element in the third metal oxide, and the content of oxygen O may be determined according to the contents of the metal elements A and C and B and stoichiometry. For example, in some examples embodiments a ratio of boron to the second metal element C may be less than or equal to one (1).

(27) The thickness of the dielectric layer **300** may be 20 Å or more and/or 100 Å or less. For example, the dielectric layer **300** may have a thickness of 25 Å or more, 30 Å or more, and/or 35 Å or more, and/or 90 Å or less, 80 Å or less, 70 Å or less, and/or 60 Å or less.

(28) The first metal oxide layer **310** may be arranged close to the lower electrode **100**, and may have a thickness of 40% or more with respect to the total thickness of the dielectric layer **300**. For example, the thickness of the first metal oxide layer **310** may be 45% or more, 50% or more, 55% or more, 60% or more, 65% or more, and/or 90% or less, and/or 85% or less, 80% or less, and/or 75% or less, with respect to the total thickness of the dielectric layer **300**. For example, the thickness of the first metal oxide layer **310** may be 10 Å or more, 15 Å or more, and/or 20 Å or more, and/or 50 Å or less, 45 Å or less, 40 Å or less, and/or 35 Å or less.

(29) Furthermore, the thickness of the second metal oxide layer **330** may be 10 Å or more, 15 Å or more, and/or 20 Å or more, and/or 50 Å or less, 45 Å or less, 40 Å or less, and/or 35 Å or less.

(30) The thickness of the third metal oxide layer **320** may be 5 Å or more, 10 Å or more, 15 Å or more, and/or 20 Å or more, and/or 50 Å or less, 45 Å or less, 40 Å or less, and/or 35 Å or less. For example, the thickness of the third metal oxide layer **320** may be 0.1 or more, 0.2 or more, and/or 0.3 or more, and/or less than 1.0, 0.9 or less, 0.8 or less, 0.7 or less, and/or 0.5 or less, with respect to the thickness of the first metal oxide layer **310**.

(31) A boundary among the first metal oxide layer **310**, the second metal oxide layer **330**, and the third metal oxide layer **320** may be unclear. For example, the boundary between the first metal oxide layer **310** and the third metal oxide layer **320**, between the second metal oxide layer **330** and the third metal oxide layer **320**, and/or among all of the first to third metal oxide layers **310**, **320**, and **330**, may be unclear. For example, when the first metal oxide layer **310**, the second metal oxide layer **330**, and the third metal oxide layer **320** are manufactured with a similar composition and/or have a small thickness, a boundary with an adjacent layer may not be clearly distinguished due to, for example, material diffusion therebetween.

(32) The dielectric layer **300** may further include a fourth metal oxide layer (not shown) including one, two, and/or more metal elements selected from Al, Mg, Si, and/or Be. The fourth metal oxide layer may or may not include boron (B). In the case wherein the fourth metal oxide layer includes B the content of B in the fourth metal oxide layer may be less than the content of B in the third metal oxide layer **320**. The fourth metal oxide layer may be between the upper electrode **200** and the second metal oxide layer **330**. The thickness of the fourth metal oxide layer may be 5 Å or more, 10 Å or more, 15 Å or more, and/or 20 Å or more, and/or 50 Å or less, 45 Å or less, 40 Å or less, and/or 35 Å or less.

(33) The capacitor **1** may further include an interface layer (not shown) between the lower electrode **100** and the dielectric layer **300**, and/or between the upper electrode **200** and the dielectric layer **300**. The interface layer may work as a barrier layer to prevent diffusion and/or movement of impurities between the lower electrode **100** and the dielectric layer **300** and/or the upper electrode **200** and the dielectric layer **300**. For example, the interface layer may prevent intrusion of some atoms, for example, nitrogen atoms, included in the upper/lower electrodes **100** and **200** into the dielectric layer **300**, and may prevent diffusion of some atoms, for example, oxygen atoms, included in the dielectric layer **300** into the upper/lower electrodes **100** and **200**. The interface layer may include a transition metal oxide having electric conductivity, and may include, for example, a metal oxide (such as a titanium oxide (TiO), a tantalum oxide (TaO), a niobium oxide (NbO), a molybdenum oxide (MoO), an iridium oxide (IrO), and the like), and/or a metal oxynitride (such as a titanium oxynitride (TiON), a tantalum oxynitride (TaON), a niobium oxynitride (NbON), a molybdenum oxynitride (MoON), and the like). The interface layer may include an oxide of metal included in the lower electrode **100** and/or the upper electrode **200**. For example, the lower electrode **100** may include a metal nitride represented by MM'N, and the interface layer between the lower electrode **100** and the dielectric layer **300** may include a metal oxynitride represented by MM'ON. The interface layer may have a thickness so as not to perform and/or perform a negligible role as a dielectric layer, and may have a thickness of, for example, about 1 Å to 10 Å.

(34) Though illustrated as including a lower electrode **100** and an upper electrode **200**, in some example embodiments, the capacitor **1** may include a plurality of internal electrodes (not illustrated) and the dielectric layer **300** may be alternatively disposed between the internal electrodes. In this case, the capacitor may be a multi-layer capacitor and may have a structure in which the plurality of internal electrodes and the dielectric layer **300** are alternately stacked. The adjacent internal electrodes may be electrically separated from one another by the dielectric layer disposed therebetween. In the multi-layer capacitor, as the internal electrodes and the dielectric layer **300** are alternately stacked, the multi-layer capacitor may act as a single unit capacitor.

(35) A capacitor according to some embodiment may include a dielectric layer including a first metal element of one, two, and/or more elements selected from Ca, Sr, Ba, Sc, Y, La, Ti, Hf, Zr, Nb, Ta, Ce, Pr, Nd, Gd, Dy, Yb, Pb, Zn, and/or Lu, a second metal element of one, two, and/or more elements selected from Al, Mg, Si, and/or Be, and B. FIG. 2 is a schematic view of a capacitor 2 according to some example embodiments. Referring to FIG. 2, the capacitor 2 may include the lower electrode 100, the upper electrode 200 arranged apart from the lower electrode 100, and a dielectric layer 301 arranged between the lower electrode 100 and the upper electrode 200. The dielectric layer 301 may include a first metal element, a second metal element, and boron (B), and the content of B may be less than or equal to the content of the second metal element.

(36) The first metal element, the second metal element, and/or B may be included in the dielectric layer 301 in appropriate contents according to a desired dielectric constant of a dielectric layer, a leakage current value of a capacitor, and the like. For example, the content of B in the dielectric layer 301 may be greater than 0.0 at %, 0.2 at % or more, 0.3 at % or more, and/or 0.5 at % or more, and/or 3.0 at % or less, 2.5 at % or less, 2.0 at % or less, and/or 1.5 at % or less, with respect to the total content of the metal elements in the dielectric layer 301. Furthermore, the content of the first metal element in the dielectric layer 301 may be 92 at % or more, 94 at % or more, and/or 95 at % or more, and/or less than 100 at %, 98 at % or less, and/or 96 at % or less, with respect to the total content of the metal elements in the dielectric layer 301. The content of the second metal element in the dielectric layer 301 may be greater than 0.0 at %, 0.3 at % or more, 0.5 at % or more, 1.0 at % or more, 1.5 at % or more, and/or 2.0 at % or more, and/or 5.0 at % or less, 4.5 at % or less, 4.0 at % or less, 3.5 at % or less, 3.0 at % or less, and/or 1.5 at % or less, with respect to the total content of the metal elements in the dielectric layer 301. The dielectric layer 301 may include a metal oxide of the first metal element, a metal oxide of the second metal element, and/or a boron oxide. For example, the dielectric layer 301 may include a metal oxide represented by  $AB_{\text{sub}.a}C_{\text{sub}.1-a}O$  wherein A may denote the first metal element, B denotes boron, and C may denote the second metal element. A may be greater than 0.00, 0.10 or more, 0.15 or more, and/or 0.20 or more, and/or 0.50 or less, and/or 0.45 or less. Furthermore, in the metal oxide represented by  $AB_{\text{sub}.a}C_{\text{sub}.1-a}O$ , the element ratio among the first metal element A, B, and the second metal element C may be determined according to the content of each metal element in the dielectric layer 301, and the content of an oxygen element may be determined according to the contents of the first metal element A, the second metal element C, and B and stoichiometry. For example, in some examples embodiments a ratio of boron to the second metal element C may be less than or equal to one (1).

(37) B may have a concentration gradation in a thickness direction of the dielectric layer 301. For example, the dielectric layer 301 may sequentially include in a thickness direction a lower surface 311 facing the lower electrode 100, an inner area 321 disposed above the lower surface 311, and an upper surface 331 disposed above the inner area 321 and facing the upper electrode 200. The boron content in the inner area 321 may be greater than that in the lower surface 311 and/or the upper surface 331. In another example, as illustrated in FIG. 3, a dielectric layer 302 may include a lower area 312 below an inner area 322 and an upper area 332 above the inner area 322, wherein the dielectric layer 302 has a certain thickness. The boron content in the inner area 322 may be greater than that in the lower area 312 and/or the upper area 332. For example, the boron content in the inner areas 321 and 322 may be 5 times or more, 10 times or more, 15 times or more, 20 times or more, 50 times or more, and/or 100 times or more than the boron contents of the lower surface/lower area 311 and 312, the upper surface/upper area 331 and 332, or all thereof. Furthermore, the content of B may have a maximum concentration (content) at a position, as measured, for example, from the lower electrode 100, of 40% or more of the thickness of the dielectric layers 301 and 302. The position where the content of B has the maximum concentration (content) may be apart from the lower electrode 100, for example, by 45% or more, 50% or more, 55% or more, and/or 60% or more, and/or 90% or less, 85% or less, 80% or less, and/or 75% or

less, with respect to thickness of each of the dielectric layers **301** and **302**.

(38) The descriptions presented above with regards to the lower electrode **100**, the upper electrode **200**, the interface layer, and the like, and the description of dielectric layer **300** may be referred to in the description of the capacitors **2** and **3**.

(39) The above-described capacitors **1**, **2**, and **3** may have a low leakage current value. For example, when a voltage of 1.0 V is applied to the capacitors **1**, **2**, and **3**, a leakage current value of  $1.0 \times 10^{-4}$  A/cm<sup>2</sup> or less,  $5.0 \times 10^{-5}$  A/cm<sup>2</sup> or less, or  $1.0 \times 10^{-5}$  A/cm<sup>2</sup> or less may be.

(40) An electrical device (e.g., a capacitor) according to some example embodiments may be manufactured by forming the lower electrode **100** on a substrate, forming the dielectric layer **300** on the lower electrode **100**, the dielectric layer **300** including the metal oxide layers **310**, **320**, and **330** having the desired compositions and thicknesses as described above, and then forming the upper electrode **200** on the dielectric layer **300**. The lower electrode **100**, the dielectric layer **300**, and the upper electrode **200** may be formed by a method that is known in the art. For example, the lower electrode **100**, the dielectric layer **300**, and the upper electrode **200** each may be independently formed through deposition methods such as atomic layer deposition (ALD), chemical vapor deposition (CVD), physical vapor deposition (PVD), sputtering, and the like. Among the methods, the ALD method has merits in that a uniform layer may be formed in an atomic unit and the method may be performed at a relatively low temperature.

(41) For example, the lower electrode **100**, the dielectric layer **300**, and the upper electrode **200** each may be independently formed by repeating one or more deposition cycles which include supplying a reactant precursor, purging the reactant precursor, supplying a reaction gas, for example, a nitrifying agent or an oxygen supply source, and purging the reaction gas.

(42) For example, when the lower and/or upper electrode **100** and **200** include a metal nitride, the lower electrode **100** and/or the upper electrode **200** including the metal nitride may be manufactured by supplying a reactant precursor (e.g., a metal precursor) and a nitrifying agent onto a substrate and/or a dielectric layer and allowing the reactant precursor and the nitrifying agent to react to each other at an appropriate temperature. A process temperature may be appropriately adjusted according to thermal stability of the metal precursor and/or the nitrifying agent, and may be 100° C. or more and/or 700° C. or less.

(43) The reactant precursor may be a metal organic compound represented by MR.sub.x and/or an elemental organic compound represented by M'R.sub.x. M or M' may be the same as the above description, R may be one or two or more of a C.sub.1 to C.sub.10 alkyl group, a C.sub.2 to C.sub.10 alkenyl group, a carbonyl group (C=O), a halide, a C.sub.6 to C.sub.10 aryl group, a C.sub.6 to C.sub.10 cycloalkyl group, a C.sub.6 to C.sub.10 cycloalkenyl group, (C=O) R (R is a hydrogen or a C.sub.1 to C.sub.10 alkyl group), a C.sub.1 to C.sub.10 alkoxy group, C.sub.1 to C.sub.10 amidinates, C.sub.1 to C.sub.10 alkylamides, C.sub.1 to C.sub.10 alkylimides, —N(Q) (Q') (Q and Q' are independently a C.sub.1 to C.sub.10 alkyl group or a hydrogen), Q(C=O)CN (Q is a hydrogen or a C.sub.1 to C.sub.10 alkyl group), and C.sub.1 to C.sub.10 β-diketonates, and x may be greater than 0 and equal to or less than 6.

(44) The reactant precursor may be a halide represented by MH.sub.y or M'H.sub.y. M or M' may be the same as the above description, and H may include one or two or more of F, Cl, Br, and I. y may be greater than 0 and equal to or less than 6.

(45) The nitrifying agent may be a reaction gas including a nitrogen element, including NH.sub.3, N.sub.2H.sub.2, N.sub.3H, and/or N.sub.2H.sub.4.

(46) After the reaction, the reactant precursor that does not react after being supplied to the substrate or the dielectric layer **300**, the unreacted reaction gas, for example, a nitrifying agent, and/or their by-products may be removed by purging. An inert gas such as Ar, He, N.sub.2, Ne, and/or the like may be used for purging.

(47) For example, as shown in FIG. **10A**, a reactant precursor (e.g., a metal precursor MR.sub.x)

may be supplied onto a substrate SU. The metal precursor may be a metal organic compound represented by MR<sub>x</sub>.

(48) As shown in FIG. 10B, the metal organic ligands may be adsorbed onto the substrate SU.

(49) The process of FIGS. 10A and 10B may be represented by Formulas 1 and 2 below:

$$xMR_4 \cdot f \cdot w \cdot d \cdot a \cdot r \cdot w \cdot xMR_4 - a + x \cdot aR \quad (1)$$

$$xMR_4 - a + x \cdot aR \cdot f \cdot w \cdot d \cdot a \cdot r \cdot w \cdot xMR_4 - a \quad (2).$$

(50) Formula (2) indicates that residual ligand components x·aR are removed by purging.

(51) Next, whether additional MR<sub>x</sub> is necessary and/or otherwise desired is determined. When additional MR<sub>x</sub> is necessary (and/or desired) the supply and purge operations may be repeated.

(52) A second source including a second reactant precursor (e.g., an elemental precursor M'R<sub>x</sub>) may be supplied, wherein M' and R<sub>x</sub> may be the same as described above. In the case wherein an atomic layer deposition (ALD) process is used in the process of supplying the second source, the ALD process may be performed at a temperature of about 100° C. to about 500° C., and the process temperature may be set in consideration of thermal stability of the metal organic ligand.

Considering that the metal organic ligand may have low thermal stability and, therefore, may be decomposed at high temperature, the ALD process may be performed at a temperature of about 400° C. or lower.

(53) Among the metal precursors provided to the reaction chamber, and the organic ligands that are not adsorbed onto the substrate SU may be removed by purging. The purging is a process of discharging organic ligands that do not participate in a reaction or organic ligands that are by-products after participating in the reaction to the outside. Inert gas such as Ar, He, Ne, or N<sub>2</sub> gas may be used for purging.

(54) The supplying the second source including the halogen compound and the purging the residual halogen compound and the by-products are shown in FIGS. 10C to 10E, and may be represented by Formulae 3-5 below:

$$yM'Cl_4 \cdot f \cdot w \cdot d \cdot a \cdot r \cdot w \cdot yM'Cl_4 - b + y \cdot bCl \quad (3)$$

$$xMR_4 - a + yM'Cl_4 - b + y \cdot bCl \cdot f \cdot w \cdot d \cdot a \cdot r \cdot w \cdot xMCl_4 - a + yM'Cl_4b + x \cdot (4 - a)R + ((y \cdot b - x \cdot (4 - a))/2)Cl_2 \quad (4)$$

$$xMCl_4 - a + yM'Cl_4 - b + x \cdot (4 - a)R + ((y \cdot b - x \cdot (4 - a))/2)Cl_2 \cdot f \cdot w \cdot d \cdot a \cdot r \cdot w \cdot xMCl_4 - a + yM'Cl_4 - b \quad (5).$$

(55) In Formulae 3-5 above, Cl is a non-limiting example as the halogen element A, and Formula (5) indicates that residual ligand components and reaction by-products are removed by purging. As shown in FIG. 10E, M supplied by the first source and M' supplied by the second source are adsorbed onto the substrate SU in a state of being bonded to the halogen element A.

(56) Next, whether additional M'Ay supply is necessary (and/or otherwise desired) is determined, and if necessary (and/or desired), the supply and purge operations may be repeated.

(57) Next, as shown in FIG. 10F, a nitrifying agent may be supplied. An ALD process may be used as the process of supplying the nitrifying agent, and may be performed at a temperature of about 100° C. to about 500° C.

(58) The nitrifying agent may be a reaction gas including a nitrogen element, including NH<sub>3</sub>, N<sub>2</sub>H<sub>2</sub>, N<sub>3</sub>H, and/or N<sub>2</sub>H<sub>4</sub>.

(59) The nitrifying agent reacts with M bonded with halogen element A and M' bonded with halogen element A, and a metal nitride film MM'N is formed on the substrate SU. Reaction by-products including halogen elements are mostly vaporized by the process temperature.

(60) The supply of the nitrifying agent and the reaction with the nitrifying agent are shown in FIGS. 10F to 10H, and may be represented by Formula 6 below:

$$xMCl_4 - a + yM'Cl_4 - b + zNH_c \cdot f \cdot w \cdot d \cdot a \cdot r \cdot w \cdot MxM'yNz + (z \cdot c)HCl + ((x \cdot (4 - a) + y \cdot (4 - b) - z \cdot c)/2)Cl_2 \quad (6).$$

(61) After the reaction, the metal precursor that does not react after being supplied to the substrate or the dielectric layer 300, the unreacted reaction gas, for example, a nitrifying agent, and/or their by-products may be removed by purging. An inert gas such as Ar, He, N<sub>2</sub>, Ne, and/or the like may be used for purging.

(62) An interface layer may be formed on the lower electrode 100 after the lower electrode 100 is

formed, and/or on the dielectric layer **300** after the dielectric layer **300** is formed. The interface layer may be formed by providing a precursor of an element that constituting the interface layer and/or supply source to the lower electrode **100** and/or to the dielectric layer **300** from a third precursor source. Alternatively, the interface layer may be formed by supplying an oxygen supply source to the lower electrode **100** to oxidize a part of a surface of the lower electrode **100**.

(63) The dielectric layers **300**, **301**, and **302** may be manufactured by providing a first metal precursor, a second metal precursor, a boron precursor, and an oxygen supply source to the lower electrode **100**, and adjusting a supply order, a supply time, a supply amount, and the like thereof, so that the dielectric layers **300**, **301**, and **302** may have desired composition, concentration, and/or thickness, for example, as discussed above. For example, the dielectric layers **300**, **301**, and **302** may be manufactured by forming the first metal oxide layer **310** by providing the first metal precursor, for example, a zirconium precursor, and an oxygen supply source to the lower electrode, forming the third metal oxide layer **320** by providing the first metal precursor, for example, the zirconium precursor, the second metal precursor, for example, an aluminum precursor, a boron precursor, and the oxygen supply source to the first metal oxide layer **310**, and forming the second metal oxide layer **330** by providing the first metal precursor, for example, the zirconium precursor, and the oxygen supply source to the third metal oxide layer **320**. The first metal precursor, the second metal precursor, the boron precursor, and/or the oxygen supply source may be simultaneously or intermittently/alternately provided to the lower electrode **100**. For example, through two or more injection holes, two or more of the first metal precursor, the second metal precursor, the boron precursor, and the oxygen supply source may be simultaneously provided to the lower electrode **100**, or the first metal precursor, the second metal precursor, the boron precursor, and the oxygen supply source may be respectively sequentially provided to the lower electrode **100**.

(64) In the dielectric layers **300**, **301**, and **302**, the composition, concentration, and/or thickness of a metal element in each layer may be changed according to a material diffusion. For example, the third metal oxide layer **320** may further comprise one, two, and/or more metal elements selected from Ca, Sr, Ba, Sc, Y, La, Ti, Hf, Zr, Nb, Ta, Ce, Pr, Nd, Gd, Dy, Yb, Pb, Zn, and/or Lu due to the material diffusion of the first metal oxide layer **310** and/or the second metal oxide layer **330** adjacent thereto, and the contents, concentrations, and/or positions in the dielectric layers of Al, Mg, Si, and Be, and/or B may vary.

(65) The metal precursors, such as the first metal precursor, the second metal precursor, and the boron (B) precursor, may be metal organic compounds that are respectively, and may be independently represented by AR.sub.x, CR.sub.x, or BR.sub.x. A may denote the first metal element, C may denote the second metal element, B may denote boron, R may be one or two or more of a C.sub.1 to C.sub.10 alkyl group, a C.sub.2 to C.sub.10 alkenyl group, a carbonyl group (C=O), a halide, a C.sub.6 to C.sub.10 aryl group, a C.sub.6 to C.sub.10 cycloalkyl group, a C.sub.6 to C.sub.10 cycloalkenyl group, (C=O) R (R is a hydrogen or a C.sub.1 to C.sub.10 alkyl group), a C.sub.1 to C.sub.10 alkoxy group, C.sub.1 to C.sub.10 amidinates, C.sub.1 to C.sub.10 alkylamides, C.sub.1 to C.sub.10 alkylimides, —N(Q)(Q') (Q and Q' are independently a C.sub.1 to C.sub.10 alkyl group or a hydrogen), Q(C=O)CN (Q is a hydrogen or a C.sub.1 to C.sub.10 alkyl group), and C.sub.1 to C.sub.10 β-diketonates, and x may be greater than 0 and equal to or less than 6. For example, A and C may be the same as the above description

(66) Ozone (O.sub.3), water (H.sub.2O), oxygen (O.sub.2), nitrous oxide (N.sub.2O), and/or plasma may be used as the oxygen supply source. The dielectric layers **300**, **301**, and **302** or the metal oxide layers **310**, **320**, and **330** may be thermally treated. For example, a thermal treatment may be performed after the dielectric layers **300**, **301**, and/or **302**; and/or after the upper electrode **200** are formed. A thermal treatment may be performed after the first metal oxide layer **310**, the third metal oxide layer **320**, and/or the second metal oxide layer **330** are formed. During a thermal treatment process, the metal element in the dielectric layers **300**, **301**, and **302** may diffuse; a part

or whole of the metal oxide in the dielectric layers **300**, **301**, and **302** and/or the metal oxide layers **310**, **320**, and **330** may be crystallized; and/or the size of a crystal grain may be increased.

(67) The thermal treatment may be performed at a temperature between 400° C. to 1100° C., but the disclosure is not limited thereto. The thermal treatment may be performed for 1 nano-second or more, 1 micro-second or more, 0.001 second or more, 0.01 second or more, 0.05 seconds or more, 0.1 second or more, 0.5 seconds or more, 1 second or more, 3 seconds or more, and/or 5 seconds or more, and/or 10 minutes or less, 5 minutes or less, 1 minute or less, and/or 30 seconds or less, but the disclosure is not limited thereto.

(68) According to another aspect, a semiconductor apparatus may be provided. The semiconductor apparatus may have memory characteristics, and may be, for example, DRAM. Furthermore, the semiconductor apparatus may include a field-effect transistor and a capacitor that are electrically connected to each other. The capacitor may be, for example, the above-described electrical device.

(69) FIG. 4A illustrates a circuit configuration of a memory cell of a memory device including a semiconductor device and a capacitor. FIG. 4B is a schematic view of a semiconductor apparatus **D1** according to some example embodiment.

(70) Referring to FIG. 4A, the semiconductor apparatus **D1** may be included in a memory device as a memory cell and may include a transistor **10** and a capacitor **1** electrically connected to, for example, a source region **11b** of the transistor **10**. The memory device may include a plurality of bit lines and a plurality of word lines, and may further include a plurality of the memory cells. Each word line may be electrically connected to a gate electrode **12b** of the transistor **10**, and each bit line may be electrically connected to a drain region **11a** of the transistor **10**. An electrode of the capacitor **1** may be connected to, for example, a voltage controller (not shown).

(71) Referring to FIG. 4B, the semiconductor apparatus **D1** may comprise the capacitor **1** and a field-effect transistor **10** electrically connected by a contact **20**. For example, one of the electrodes **100** and **200** of the capacitor **1** and one of a source and a drain **11a** and **11b** of the field-effect transistor **10** may be electrically connected by the contact **20**. Although FIG. 4B illustrates an example of the semiconductor apparatus **D1** including the capacitor **1** of FIG. 1, the capacitors **2** and **3** of FIGS. 2 and 3 may be included in the semiconductor apparatus **D1**. Furthermore, though FIG. 4B illustrates an example of the semiconductor apparatus **D1** wherein the contact **20** is illustrated as a separate element from the source **11a** and the electrode **100** the semiconductor apparatus **D1** is not so limited. For example, the contact **20**, source **11a**, and electrode **100** may comprise a shared conductive material layer and/or a series of electrically connected conductive components.

(72) The field-effect transistor **10** may include a substrate **11** and a gate electrode **12b** facing a channel **11c**. A gate insulating layer **12a** may be further provided between the substrate **11** and the gate electrode **12b**.

(73) The substrate **11** may include a semiconductor material. The substrate **11** may include a semiconductor material, for example, Si, Ge, SiGe, SiC, GaAs, InAs, InP, and/or the like, and may be applied in various forms such as silicon on insulator (SOI), and the like.

(74) The substrate **11** may include the channel **11c** electrically connected to the source **11a**, and the drain **11b**. The source **11a** may be electrically connected to and/or in contact with an end portion of one side of the channel **11c**, and the drain **11b** may be electrically connected to and/or in contact with an end portion of the other side of the channel **11c**. For example, the channel **11c** may be a substrate area between the source **11a** and the drain **11b** in the substrate **11** configured to form a channel connection between the source **11a** and the drain **11b** when an electrical charge is applied to the gate electrode **12b**.

(75) The source **11a**, the drain **11b**, and the channel **11c** each may be independently formed by injecting impurities in different areas of the substrate **11**. In this case, the source **11a**, the channel **11c**, and the drain **11b** may include a substrate material as a base material.

(76) Furthermore, the source **11a** and the drain **11b** may be formed of a conductive material, and

for example, each of the source **11a** and the drain **11b** may independently include metal, a metal compound, and/or a conductive polymer.

(77) The channel **11c** may include a separate material layer (thin film) (not shown). In this case, for example, the channel **11c** may include a semiconductor material such as Si, Ge, SiGe, group III-V semiconductors, and the like, an oxide semiconductor, a nitride semiconductor, an oxynitride semiconductor, a two-dimensional material (2D material), a quantum dot (QD), and/or an organic semiconductor. For example, the oxide semiconductor may include InGaZnO, and the like, the 2D material may include transition metal dichalcogenide (TMD) and/or graphene, and the QD may include a colloidal QD or nanocrystal structure.

(78) The gate electrode **12b** may be disposed on the substrate **11**, and spaced apart from the substrate **11** and facing the channel **11c**. The gate electrode **12b** may have a conductivity of 1 Mohm/square or less. The gate electrode **12b** may include metal, a metal nitride, a metal carbide, and/or polysilicon. For example, the metal may include Al, W, Mo, Ti, and/or Ta, and a metal nitride film may include a titanium nitride film (TiN film) and/or a tantalum nitride film (Ta<sub>2</sub>N film). The metal carbide may include a metal carbide doped (or contained) with aluminum and/or silicon, and may include TiAlC, TaAlC, TiSiC, or TaSiC, as a detailed example. The gate electrode **12b** may include a block material and/or a plurality of stacked materials, for example, a stacking structure of a metal nitride layer/a metal layer such as TiN/Al, and the like, and/or a stacking structure of a metal nitride layer/a metal carbide layer/a metal layer such as TiN/TiAlC/W. The gate electrode **12b** may include TiN and/or Mo in the form in which the above example is variously deformed.

(79) The gate insulating layer **12a** may be further provided between the substrate **11** and the gate electrode **12b**. The gate insulating layer **12a** may include a paraelectric material and/or a high dielectric material, and may have a dielectric constant of 20 to 70. The gate insulating layer **12a** may include a silicon oxide, a silicon nitride, an aluminum oxide, a hafnium oxide, a zirconium oxide, and the like, or may include a 2D insulator such as a hexagonal boron nitride (h-BN). For example, the gate insulating layer **12a** may include SiO<sub>2</sub>, SiN<sub>x</sub>, and the like, and HfO<sub>2</sub>, HfSiO<sub>4</sub>, La<sub>2</sub>O<sub>3</sub>, LaAlO<sub>3</sub>, ZrO<sub>2</sub>, HfZrO<sub>2</sub>, ZrSiO<sub>4</sub>, Ta<sub>2</sub>O<sub>5</sub>, TiO<sub>2</sub>, SrTiO<sub>3</sub>, Y<sub>2</sub>O<sub>3</sub>, Al<sub>2</sub>O<sub>3</sub>, PbSc<sub>0.5</sub>Ta<sub>0.5</sub>O<sub>3</sub>, PbZnNbO<sub>3</sub>, and the like. Furthermore, the gate insulating layer **12a** may include a metal oxynitride such as AlON, ZrON, HfON, LaON, YON, and the like, a silicate such as ZrSiON, HfSiON, YSiON, LaSiON, and the like, or an aluminate such as ZrAlON, HfAlON, and the like. Furthermore, the gate insulating layer **12a** may include the above-described dielectric layers **300**, **301**, and **302**. The gate insulating layer **12a** may constitute a gate stack with the gate electrode **12b**.

(80) The contact **20** may include a conductive material, for example, W, Cu, AL, polysilicon, and the like.

(81) The arrangement of the capacitor **1** and the field-effect transistor **10** may be variously changed. For example, the capacitor **1** may be disposed on the substrate **11**, and/or may comprise a structure embedded in the substrate **11**.

(82) For example, referring to FIG. **11**, on a semiconductor substrate **1100**, a device isolation region may be defined with a field oxide film **821**, and a gate electrode **110** and source/drain impurity regions **130** and **120** may be formed in the device isolation region. A dielectric layer **410** may be formed between the gate electrode **110** and the semiconductor substrate **1100**. An oxide film may be formed as an interlayer insulating film **824**. A region not to be a trench may be capped with a trench buffer layer, and a part of the source region **130** may be open to form a contact portion.

(83) A trench may be formed in a sidewall of the interlayer insulating film **824**, and a sidewall oxide film **825** may be formed over the entire sidewall of the trench. The sidewall oxide film **825** may compensate for damage in the semiconductor substrate caused by etching to form the trench, and may serve as a dielectric film between the semiconductor substrate **1100** and a first electrode



**600**. A sidewall portion of part of the source region **130**, except for the other part of the source region near the gate electrode **110**, may be entirely exposed.

(84) A PN junction (not illustrated) may be formed in the sidewall portion of the source region by impurity implantation. The trench may be formed in the source region **130**. A sidewall of the trench near the gate may directly contact the source region **130**, and the PN junction may be formed by additional impurity implantation into the source region.

(85) The first electrode **600** may be formed on part of the interlayer insulating film **824**, the exposed source region **130**, and the surface of the sidewall oxide film **825** in the trench. The first electrode **600** may contact the entire source region **130** in contact with the upper sidewall of the trench, in addition to the part of the source region **130** near the gate electrode **110**. Next, a dielectric layer **300** may be formed as a capacity dielectric film along the upper surface of the first electrode **600**, and a second electrode **700** may be formed thereon, thereby completing a trench capacitor type DRAM. Though illustrated as including the dielectric layer **300** in only the capacity dielectric film, any of the gate insulating film **410**, capacity dielectric film, and/or the interlayer insulating film **824**, for example, may include an embodiment of the dielectric layer **300**, **301**, and/or **302**, as described above.

(86) Although FIGS. **4A** and **4B** schematically illustrates the semiconductor apparatus **D1** having one capacitor **1** and one field-effect transistor **10**, as illustrated in FIG. **5**, a semiconductor apparatus **D10** may have a structure in which a plurality of capacitors and a plurality of field-effect transistors are repeatedly arranged. Referring to FIG. **5**, the semiconductor apparatus **D10** may include a field-effect transistor including the substrate **11'** having a source, a drain, and a channel and the gate stack **12**, the contact structure **20'** arranged on the substrate **11'** not to overlap the gate stack **12**, and a the capacitor **1'** disposed on the contact structure **20'**, and may further include the bit line structure **13** that electrically connects the field-effect transistors. Although FIG. **5** illustrates an example of the semiconductor apparatus **D10** in which the contact structure **20'** and the capacitor **1'** both are repeatedly arranged in an X direction and a Y direction, the disclosure is not limited thereto. For example, the contact structure **20'** may be arranged in the X direction and the Y direction, and the capacitor **1'** may be arranged in a hexagonal shape such as a honeycomb structure.

(87) FIG. **6** is a cross-sectional view taken along line A-A' of the semiconductor apparatus **D10** of FIG. **5**. Referring to FIG. **6**, the substrate **11'** may have a shallow trench isolation (STI) structure including a device separation film **14**. The device separation film **14** may be a single layer including one type of insulating film or a multiple layer including a combination of two types or more of insulating films. The device separation film **14** may include a device separation trench **14T** in the substrate **11'**, and the device separation trench **14T** may be filled with an insulating material. The insulating material may include a low dielectric constant material, and may be, for example, fluoride silicate glass (FSG), undoped silicate glass (USG), boro-phospho-silicate glass (BPSG), phospho-silicate glass (PSG), a silicone dielectric (e.g., flowable oxide (FOX)), plasma enhanced tetra-ethyl-ortho-silicate (PE-TEOS), and/or polysilazane (e.g., tonen silazane (TOSZ)), but the disclosure is not limited thereto.

(88) Furthermore, the substrate **11'** may further include an active area **AC** defined by the device separation film **14**, and a gate line trench **12T** that is parallel to an upper surface of the substrate **11'** and extends in the X direction. The active area **AC** may have a relatively long island shape having a minor axis and a major axis. The major axis of the active area **AC**, as illustrated in FIG. **5**, may be arranged in a direction **D3** parallel to the upper surface of the substrate **11'**. The gate line trench **12T** may be disposed to a certain depth from the upper surface of the substrate **11'** to cross the active area **AC** or in the active area **AC**. The gate line trench **12T** may be disposed in the device separation trench **14T**, and the gate line trench **12T** in the device separation trench **14T** may have a bottom surface lower than the gate line trench **12T** of the active area **AC**.

(89) A first source/drain **11'ab** and a second source/drain **11''ab** may be disposed in an upper

portion of the active area AC located at both sides of the gate line trench 12T.

(90) The gate stack 12 may be disposed in the gate line trench 12T. For example, the gate insulating layer 12a, the gate electrode 12b, and a gate capping layer 12c may be sequentially disposed in the gate line trench 12T. The gate insulating layer 12a and the gate electrode 12b may refer to the above-described content, and the gate capping layer 12c may include a silicon oxide, a silicon oxynitride, and/or a silicon nitride. The gate capping layer 12c may be disposed on the gate electrode 12b to fill a remaining portion of the gate line trench 12T.

(91) Furthermore, the bit line structure 13 may be disposed on the first source/drain 11'ab. The bit line structure 13 may be disposed parallel to the upper surface of the substrate 11' and extending in the Y direction. The bit line structure 13 may be electrically connected to the first source/drain 11'ab, and may sequentially include a bit line contact 13a, a bit line 13b, and a bit line capping layer 13c on the substrate 11'. For example, the bit line contact 13a may include polysilicon, the bit line 13b may include a metal material, and the bit line capping layer 13c may include an insulating material such as a silicon nitride, a silicon oxynitride, and the like. Although FIG. 6 illustrates an example in which the bit line contact 13a has a bottom surface at the same level as the upper surface of the substrate 11', the bit line contact 13a may extend to the inside of a recess (not shown) formed to a certain depth from the upper surface of the substrate 11', so that the bottom surface of the bit line contact 13a may be lower than the upper surface of the substrate 11'.

(92) Optionally, the bit line structure 13 may include a bit line intermediate layer (not shown) between the bit line contact 13a and the bit line 13b. The bit line intermediate layer may include a metal silicide such as tungsten silicide, and/or a metal nitride such as a tungsten nitride. Furthermore, a bit line spacer (not shown) may be further formed on a sidewall of the bit line structure 13. The bit line spacer may have a single layer structure or a multilayer structure, and may include an insulating material such as a silicon oxide, a silicon oxynitride, or a silicon nitride. Furthermore, the bit line spacer may further include an air space (not shown).

(93) The contact structure 20' may be disposed on the second source/drain 11''ab. The contact structure 20' and the bit line structure 13 may be disposed on a different source/drain on the substrate 11'. The contact structure 20' may be a structure in which a lower contact pattern (not shown), a metal silicide layer (not shown), and an upper contact pattern (not shown) are sequentially stacked on the second source/drain 11''ab. Furthermore, the contact structure 20' may further include a barrier layer (not shown) surrounding a side surface and a bottom surface of the upper contact pattern. For example, the lower contact pattern may include polysilicon, the upper contact pattern may include a metal material, and the barrier layer may include a metal nitride having conductivity.

(94) The capacitor 1' may be electrically connected to the contact structure 20' and disposed on the substrate 11'. In detail, the capacitor 1' may include the lower electrode 100 electrically connected to the contact structure 20', the dielectric layer 300 disposed on the lower electrode 100, and the upper electrode 200 disposed on the dielectric layer 300. The dielectric layer 300 may be disposed on the lower electrode 100 to be parallel to a surface of the lower electrode 100.

(95) The interlayer insulating layer 15 may be further disposed between the capacitor 1' and the substrate 11'. The interlayer insulating layer 15 may be disposed in a space between the capacitor 1' and the substrate 11' where no other structure is disposed. In detail, the interlayer insulating layer 15 may be disposed to cover a wiring and/or electrode structure of the bit line structure 13, the contact structure 20', the gate stack 12, and the like on the substrate 11'. For example, the interlayer insulating layer 15 may surround a wall of the contact structure 20'. The interlayer insulating layer 15 may include a first interlayer insulating layer 15a surrounding the bit line contact 13a and a second interlayer insulating layer 15b covering side surfaces and/or upper surfaces of the bit line 13b and the bit line capping layer 13c.

(96) The lower electrode 100 of the capacitor 1' may be disposed on the interlayer insulating layer 15, in detail, on the second interlayer insulating layer 15b. Furthermore, when a plurality of

capacitors **1'** are disposed, a bottom surface of a plurality of lower electrodes **100** may be separated by an etching stop layer **16**. The etching stop layer **16** may include an opening portion **16T**, and the bottom surface of the lower electrode **100** of the capacitor **1'** may be disposed in the opening portion **16T**.

(97) The lower electrode **100** may have a cylinder shape and/or a cup shape with a closed bottom surface, as illustrated in FIG. **6**. In another example, the lower electrode **100** may have a pillar shape such as a circular pillar, a rectangular pillar, and/or a polygonal pillar extending in a vertical direction (Z direction), as illustrated in FIGS. **6** and **7**.

(98) Furthermore, the capacitor **1'** may further include a support (not shown) to prevent the tilting and/or collapsing of the lower electrode **100**, and the support may be disposed on a sidewall of the lower electrode **100**.

(99) The semiconductor apparatuses **D20** and **D30** may be manufactured by referring to a typical method known in the art. For example, the semiconductor apparatuses **D20** and **D30** may be manufactured including operations i) to xvi) below. i) Forming the device separation trench **14T** in the substrate **11'**, and forming the device separation film **14** in the device separation trench **14T** (defining the active area AC of the substrate **11'** by the device separation film **14** and/or the device separation trench **14T**), ii) Filling the inside of the device separation trench **14T** with an insulating material, iii) Forming the first source/drain **11'ab** and the second source/drain **11''ab** in an upper area of the active area AC by injecting impurity ions into the substrate **11'**, iv) Forming the gate line trench **12T** in the substrate **11'**, v) Forming the gate insulating layer **12a**, the gate electrode **12b**, and the gate capping layer **12c** in the gate line trench **12T**, vi) Forming the first interlayer insulating layer **15a** on the substrate **11'**, and forming an opening portion (not shown) that exposes the upper surface of the first source/drain **11'ab**, vii) Forming, on the opening portion of vi), the bit line structure **13** electrically connected to the first source/drain **11'ab**, viii) Forming the second interlayer insulating layer **15b** that covers an upper surface and a side surface of the bit line structure **13**, ix) forming an opening portion (not shown) in the first and second interlayer insulating layers **15a** and **15b** to expose an upper surface of the second source/drain **11''ab**, x) Forming, on the opening portion of ix), the contact structure **20'** that is electrically connected to the second source/drain **11''ab**, xi) Forming the etching stop layer **16** and a mold layer (not shown) on the second interlayer insulating layer **15b** and the contact structure **20'**, xii) Forming an opening portion (not shown) in the etching stop layer **16** and a mold layer (not shown) to expose an upper surface of the contact structure **20'**, xiii) Forming the lower electrode **100** to cover an inner wall of the opening portion of xii) (to cover a bottom surface and a side surface of the opening portion of xii). xiv) Removing the mold layer, xv) Forming the dielectric layer **300** on the lower electrode **100**, and xvi) Forming the upper electrode **200** on the dielectric layer **300**.

(100) However, the types and/or orders of the above-described operations are not so limited, and may be appropriately adjusted and/or some operations may be omitted and/or added. Furthermore, in each operation, a deposition process, a patterning process, an etching process, and the like, which are known to the art, may be used for forming constituent elements. For example, during the forming of an electrode, an etchback process may be employed. In operation v), the gate electrode **12b** may be formed by forming a conductive layer on the gate insulating layer **12a** and then removing an upper portion of the conductive layer to a certain height through an etchback process. Furthermore, in operation xiii), the lower electrode **100** may be manufactured by forming an electrode to cover all of an upper surface of the mold layer and the bottom surface and the side surface of the opening portion, and then removing a part of the electrode on the upper surface of the mold layer by an etchback process, thereby manufacturing a structure having a plurality of lower electrodes **100**. In another example, a planarization process may be employed. For example, in operation v), the gate capping layer **12c** may be formed by filling the remaining portion of the gate line trench **12T** with an insulating material, and then polarizing the insulating material until the upper surface of the substrate **11'** is exposed.

(101) The semiconductor device and the semiconductor apparatus may be applied to various electronic apparatuses. For example, the above-described semiconductor device and/or semiconductor apparatus may be applied to various electronic apparatuses as a logic device and/or a memory device. For example, the semiconductor device and the semiconductor apparatus may be used for an arithmetic operation, program execution, temporary data maintenance, and the like, in electronic apparatuses such as mobile devices, computers, notebook computers, sensors, network apparatuses, neuromorphic devices, and the like. The electronic device according to some embodiments and the semiconductor apparatus may be useful for electronic apparatuses in which a data transfer amount is large and data transmission is continuously performed.

(102) FIGS. 8 and 9 are conceptual views schematically showing device architectures **1000** applicable to an electronic apparatus, according to some embodiments.

(103) Referring to FIG. 8, the electronic device architecture **1000** may include a memory unit **1010**, an arithmetic logic unit (ALU) **1020** and a control unit **1030**. The memory unit **1010**, the ALU **1020**, and the control unit **1030** may be electrically connected to one another. For example, the electronic device architecture **1000** may be implemented as one chip including the memory unit **1010**, the ALU **1020**, and the control unit **1030**. The memory unit **1010**, the ALU **1020**, and the control unit **1030** may be configured to directly communicate with one another by, for example, mutual connections to a metal line in on-chip. The memory unit **1010**, the ALU **1020**, and the control unit **1030** may be monolithically integrated on one substrate, thereby forming one chip. An input/output device **2000** may be connected to the electronic device architecture (chip) **1000**. Furthermore, the memory unit **1010** may include both main memory and cache memory. The electronic device architecture (chip) **1000** may be an on-chip memory processing unit.

(104) The memory unit **1010**, the ALU **1020**, and/or the control unit **1030** each may independently include the above-described semiconductor device. Referring to FIG. 9, a cache memory **1510**, an ALU **1520**, and a control unit **1530** may constitute a central processing unit (CPU) **1500**, and a cache memory **1510** may include static random access memory (SRAM). Aside from the CPU **1500**, a main memory **1600** and an auxiliary storage **1700** may be provided. The main memory **1600** may include dynamic random access memory (DRAM) and may include the above-described semiconductor device.

(105) In some cases, the electronic device architecture may be implemented such that, without distinction of sub-units, computing unit devices and memory unit devices are mutually adjacent to each other in one chip.

(106) The technical contents of the semiconductor device may be described in detail below through the following embodiments. However, the following embodiments are merely for the purpose of explanation, and should not limit the scope of rights.

#### Embodiment 1

(107) A lower electrode is formed by a DC sputter or an ALD method.

(108) A dielectric layer is formed on the lower electrode through the ALD method. For example, after forming a first metal oxide layer including ZrO.sub.2, a third metal oxide layer including Zr, Al, and B is formed on the first metal oxide layer, and a second metal oxide layer including ZrO.sub.2 is formed on the third metal oxide layer. Consequently, the dielectric layer may have a concentration gradation of a boron content of an inner area being greater than boron contents of an upper surface and a lower surface. Furthermore, the amounts of a Zr precursor, an Al precursor, and a B precursor are adjusted during the forming of the dielectric layer such that a content ratio of a B element to an Al element in the third metal oxide layer is 1.0, a content ratio of the B element to a total of metal elements in the third metal oxide layer is 5.0 at %, or a content ratio of the B element to a total of metal elements in the dielectric layer is 2.0 at %.

(109) An upper electrode is formed on the dielectric layer by the DC sputter or the ALD method.

(110) The layers and the electrodes formed as above undergo a rapid thermal annealing (RTA) at a temperature between 400° C. to 1000° C., thereby manufacturing a capacitor.

#### Embodiments 2 and 3

(111) A capacitor is manufactured in the same method as the method of embodiment 1, except that a content ratio between metal elements in the third metal oxide layer and/or a content ratio between metal elements in the dielectric is adjusted, during the forming of the third metal oxide layer, by changing the amounts of the Zr precursor, the Al precursor, and the B precursor as depicted in Table 1.

#### Comparative Example 1

(112) A capacitor is manufactured in the same method as that of embodiment 1, except that, during the forming of the third metal oxide layer, the third metal oxide layer including Zr and Al is formed without using the B precursor.

#### Comparative Example 2

(113) A capacitor is manufactured in the same method as that of embodiment 1, except that, during the forming of the third metal oxide layer, the third metal oxide layer including Zr and B is formed without using the Al precursor.

#### Electrical Property Evaluation 1

(114) A capacitance is measured by applying 1.0 V to the capacitor manufactured by the methods of embodiments 1 to 3 and comparative examples 1 and 2, and is normalized with respect to the capacitance of the capacitor of comparative example 1. Each capacitance of the capacitor is written in Table 1.

(115) Furthermore, a leakage current value is measured by applying 1.0 V to the capacitors of embodiments 1 to 3 and comparative examples 1 and 2, and the measured value is written in Table 1.

(116) Referring to Table 1, the capacitors of embodiment 1 to 3 including Al and B in the dielectric layer shows 15% or more high capacitance compared to the capacitor of comparative example 1 that does not include the oxide layer including boron (B). Furthermore, while the capacitors of embodiment 1 to 3 show a leakage current value of  $1.0 \times 10^{-5}$  A/cm<sup>2</sup> or less, the capacitor of comparative example 2 shows a leakage current value of greater than  $1.0 \times 10^{-4}$  A/cm<sup>2</sup>. In other words, it may be checked that the capacitors of embodiment 1 to 3 have a high capacitance and simultaneously a low leakage current value.

(117) TABLE-US-00001 TABLE 1 In third metal In dielectric oxide layer layer Leakage B/Al element B element current ratio content (at %) Capacitance (A/cm<sup>2</sup>) Embodiment 1 1 2 115% 1.7E-6 Embodiment 2 0.67 1.3 116% 1.6E-6 Embodiment 3 0.33 0.7 118% 1.6E-6 Comparative 0 0 100% 1.5E-6 example 1 Comparative — 3.4 130% 2.0E-4 example 2

#### Embodiments 4 and 5

(118) A capacitor is manufactured in the same method as that of embodiment 1, except that, by adjusting the thicknesses of the first metal oxide layer and the second metal oxide layer, the third metal oxide layer is located at a position of 40% or more of the total thickness of the dielectric layer from the lower electrode (embodiment 4), or at a position of less than 40% of the total thickness of the dielectric layer from the lower electrode (embodiment 5). Consequently, in the dielectric layer of embodiment 4, B may have the maximum concentration at a position at 40% or more of the thickness of the dielectric layer, as measured from the lower electrode, and in the dielectric layer of embodiment 5, B may have the maximum concentration at a position away from the lower electrode by less than 40% of the thickness of the dielectric layer.

#### Electrical Property Evaluation 2

(119) A capacitance by applying 1.0 V to the capacitors of embodiments 4 and 5 and a leakage current value when 1.0 V is applied are measured and written in Table 2. The capacitance is normalized based on the capacitance of the capacitor of embodiment 4.

(120) Referring to Table 2, although the capacitors of embodiments 4 and 5 both have a low level leakage current value, the capacitor of embodiment 5, in which the third metal oxide layer is located at a position less than 40% of the total thickness of the dielectric layer, shows a lower

capacitance than embodiment 4.

(121) TABLE-US-00002 TABLE 2 Position of third Leakage current metal oxide layer Capacitance value (A/cm.sup.2) Embodiment 4 40% or more of 100%  $1.7 \times 10^{-6}$  total thickness of dielectric layer Embodiment 5 Less than 40% of 68%  $1.7 \times 10^{-7}$  total thickness of dielectric layer

(122) According to the above embodiments, an electrical device having a high capacitance and a superior leakage current blocking/reduction characteristics, and a semiconductor apparatus including the electrical device, may be provided. The electrical device may implement improved degree of integration, and may contribute to the miniaturization of an electronic apparatus.

(123) It should be understood that embodiments described herein should be considered in a descriptive sense only and not for purposes of limitation. Descriptions of features or aspects within each embodiment should typically be considered as available for other similar features or aspects in other embodiments. While one or more embodiments have been described with reference to the figures, it will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope as defined by the following claims.

## Claims

1. An electrical device comprising: a lower electrode; an upper electrode isolated from direct contact with the lower electrode; and a dielectric layer between the lower electrode and the upper electrode, wherein the dielectric layer comprises a first metal oxide area, a second metal oxide area, and a third metal oxide area between the first metal oxide area and the second metal oxide area, wherein the third metal oxide area comprises boron (B) and at least one of Al, Mg, Si, or Be, such that a concentration of the boron in the dielectric layer, as measured from the lower electrode, at a position between 40% to 90% a thickness of the dielectric layer is greater than the concentration of the boron in the dielectric layer lower than the position between 40% to 90% the thickness of the dielectric layer.
2. The electrical device of claim 1, wherein the first metal oxide area, the third metal oxide area, and the second metal oxide area are sequentially arranged in a thickness direction of the dielectric layer.
3. The electrical device of claim 1, wherein each of the first metal oxide area and the second metal oxide area has a dielectric constant of 20 or more and 70 or less.
4. The electrical device of claim 1, wherein each of the first metal oxide area and the second metal oxide area independently comprise an oxide including at least one of Ca, Sr, Ba, Sc, Y, La, Ti, Hf, Zr, Nb, Ta, Ce, Pr, Nd, Gd, Dy, Yb, Pb, Zn, or Lu.
5. The electrical device of claim 1, wherein a boron content in the third metal oxide area is greater than a boron content in each of the first and second metal oxide areas.
6. The electrical device of claim 1, wherein a boron content in the third metal oxide area is less than or equal to a content of metal elements comprising the third metal oxide area.
7. The electrical device of claim 1, wherein the first metal oxide area comprises a first metal oxide layer, the second metal oxide area comprises a second metal oxide layer, the third metal oxide area includes a third metal oxide layer, and the concentration of the boron included in the dielectric layer changes across an interface between third metal oxide layer and the first metal oxide layer.
8. The electrical device of claim 1, wherein the third metal oxide area comprises a metal oxide represented by  $AB_{\text{sub.a}}C_{\text{sub.1-a}}O$ , wherein A is at least one of Ca, Sr, Ba, Sc, Y, La, Ti, Hf, Zr, Nb, Ta, Ce, Pr, Nd, Gd, Dy, Yb, Pb, Zn, or Lu, B is boron (B), C is at least one of Al, Mg, Si, and Be, and a is greater than 0.00 and 0.50 or less.
9. The electrical device of claim 1, wherein the first metal oxide area is adjacent to the lower electrode, and a thickness of the first metal oxide area is 40% or more of a total thickness of the

dielectric layer.

10. The electrical device of claim 1, wherein a ratio of a thickness of the third metal oxide area, with respect to a thickness of the first metal oxide area, is 0.3 or more and less than 1.0.

11. The electrical device of claim 1, wherein the dielectric layer comprises a fourth metal oxide area between the upper electrode and the second metal oxide area, the fourth metal oxide area comprising at least one of Al, Mg, Si, or Be.

12. The electrical device of claim 11, wherein a boron content in the fourth metal oxide area is less than a content of boron in the third metal oxide area.

13. The electrical device of claim 1, wherein the dielectric layer is configured such that, when a voltage of 1.0 V is applied, a leakage current value is  $1.0 \times 10^{-4}$  A/cm<sup>2</sup> or less.

14. The electrical device of claim 1, wherein at least one of the upper electrode and the lower electrode comprises a metal nitride represented by MM'N, wherein M is a metal element, M' is a doping element, and N is nitrogen.

15. The electrical device of claim 1, further comprising: an interface layer between the dielectric layer and at least one of the upper electrode and the lower electrode.

16. A semiconductor apparatus, the semiconductor apparatus comprising: the electrical device according to claim 1; and a field-effect transistor electrically connected to the electrical device.

17. The semiconductor apparatus of claim 16, wherein the field-effect transistor comprises: a source and a drain; a semiconductor channel connecting the source and the drain; a gate dielectric layer on the semiconductor channel; and a gate electrode on the gate dielectric layer.

18. A dynamic random access memory (DRAM) comprising: a plurality of DRAM cells, each of the DRAM cells including the electrical device according to claim 1.

19. A capacitor comprising: a lower electrode; an upper electrode spaced apart from the lower electrode; and a dielectric layer between the lower electrode and the upper electrode such that the dielectric layer electrically insulates the lower electrode from the upper electrode, wherein the dielectric layer includes a metal oxide, Boron (B) and at least one of Al, Mg, Si, and Be, wherein, at a position between 40% to 90%, a concentration of the boron in the dielectric layer changes such that the concentration above the position is greater than the concentration below the position.

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